JULIANNA C. HSING, MS

PhD Candidate | Epidemiology & Clinical Research | Stanford University

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Summary

PhD Candidate in the Department of Epidemiology & Population Health at Stanford School of Medicine, specializing in **digital health**, **sleep health**, **resilience**, **gut microbiome**, and **big data**. My research leverages smartphone sensor data and advanced methods for longitudinal analysis to understand how stress and screen time contribute to sleep behaviors in young adults. Bridging science and art, I bring strong expertise in **visual storytelling, information design, and science communication**—transforming complex, high-dimensional data into clear, impactful narratives that make health research accessible to diverse audiences.

Work Experience

PhD Student Researcher

2023-Present   digital health   sleep health   big data

Dept. of Epidemiology & Population Health | Stanford University School of Medicine

- Designed and built digital pipelines to process and visualize over 10 billion rows of smartphone sensor data (e.g., accelerometer, GPS, screen state) into meaningful behavioral insights
- Analyzed over 20 complex digital biomarkers that provide insights about stress, sleep, and movement patterns in young adults
- Presented scientific findings to 1000+ attendees at a national scientific conference, emphasizing clarity and design in communicating research to diverse audiences

Research Data Analyst

2018-2023   digital health   resilience   longitudinal analysis

Dept. of Pediatrics | Stanford University School of Medicine

- Co-authored 10+ peer-reviewed articles (4 first-authored), each featuring visualizations that distilled technical results into compelling stories. Topics included: child health, mobile health, COVID-19, and resilience
- Supervised 2–4 research assistants weekly, standardizing workflows and introducing best practices for reproducible research
- Contributed to the design and testing of PretermConnect, a mobile health app reaching hundreds of high-risk families. Also led the statistical analysis for this project, transforming raw data into graphics used by researchers, clinicians, and policymakers

Undergraduate Researcher

2014-2017   gut microbiome   big data

Dept. of Ecology and Evolutionary Biology | Princeton University

- Conducted bioinformatics analyses on hundreds of mammalian gut microbiome samples, producing comparative visualizations of microbial diversity
- Conducted DNA-based molecular lab work on large mammalian herbivore fecal samples from Kenya, Africa
- Collaborated with an interdisciplinary team, sharpening early skills in visual communication and scientific storytelling

Education

Stanford University

PhD, Epidemiology & Clinical Research (2028 *Expected*)

MS, Epidemiology & Clinical Research (2021)

University of Pennsylvania

Post-baccalaureate, Pre-Health (2017)

Princeton University

BA, Ecology and Evolutionary Biology (2016)

Technical Skills

Programming

R (12 years), Python (5 years), SQL (3 years), D3.js (1 year)

Creative Software

Procreate (5 years), Adobe Illustrator (3 years), Adobe InDesign (3 years), Flourish (2 years), Adobe Photoshop (basic), Figma (basic)

Language

Mandarin Chinese (Near-native speaking; basic reading and writing proficiency)

Extracurricular & Interests

Resident Artist, Stanford WELL for Life Study

Sports: climbing, taekwondo (Black Belt)
Music: violin (25 years), guitar (8 years)
Art: information design, printmaking (letterpress, risograph, linocut), ceramics

Certifications

Data Science and Machine Learning: Making Data-Driven Decisions (2022)

MIT Institute for Data, Systems, and Society